Landing on the Wrong Altimeter Setting: A Pre-registered Study of Business Jets and Airline Jets in Open ADS-B Data

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Abstract

An aircraft that lands with the wrong altimeter setting flies its barometric altitudes at the wrong height with no barometric indication of it. Two European studies have measured how often this happens; no published measurement was found for the United States, or one that compares types of operator. Modern aircraft broadcast the pressure setting selected on board. This study compared that broadcast setting with the consensus of other aircraft landing at the same airport within 30 minutes, at 119 US and European airports, in two pre-registered samples of 30 days in 2026 (about 714,000 and 730,000 landings). In the first sample business jets landed with a setting at least 2 hPa from the consensus more often than airline jets at the same airport on the same day (Mantel-Haenszel odds ratio 3.4). A second hypothesis, about Europe against the United States, was not supported, and review showed that the registered measure picked up a spurious switch to the standard setting in the final seconds on some European airliners. A corrected measure was then fixed in advance and tested on the second sample. The business-jet result replicated: odds ratio 4.0 (95% interval 3.5 to 4.7; 2.7 to 5.4 when whole operators are resampled), and 13.9 (7.9 to 23.7) for errors of 5 hPa or more. In the second sample the setting was wrong by 2 hPa or more on 2.0 per 1,000 landings, by 5 hPa or more on 0.19, and by 10 hPa or more (about 280 ft) on 0.06. The consensus was within 1 hPa of the airport weather report for 97% of landings, and the results were the same with the weather report as the reference. Nine in ten of the discrepancies are of 2 to 5 hPa and may not be errors in the operational sense. About half of the large errors recorded in Europe came from a few older aircraft that supply the value by a second route and show the standard setting on most landings; with those removed, errors of 5 hPa or more were two to three times as frequent in the United States as in Europe. Which altimeter the broadcast reflects is not defined, so on some business jets a wrong broadcast may not mean the crew flew on a wrong setting. Review of this draft has been by AI reviewers only.

1 Introduction

On 23 May 2022 an Airbus A320 on a satellite-based approach to Paris-Charles de Gaulle descended to about six feet above the ground about 0.9 NM short of the runway. The controller had passed an altimeter setting 10 hPa too high, the crew set it, and every barometric indication on board looked normal [1]. The European Union Aviation Safety Agency has issued and revised a safety bulletin on incorrect altimeter settings [2].

Two published measurements of how often aircraft land on a wrong setting were found, both in Europe. The French civil aviation safety directorate compared the setting downlinked to radar with the setting displayed to controllers for 1,694,266 landings at six French airports

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from 2019 to 2023. It found a difference of 2 hPa or more at the runway threshold on 2,269 landings (about 1.3 per 1,000) and of 10 hPa or more on 387 (about 0.23 per 1,000). About 27% of the inconsistencies had the standard setting on board. The rate at Le Bourget, a business-aviation airport, was three times the rate at Charles de Gaulle, which the report attributed to "very different fleets" without further analysis [3]. A EUROCONTROL study built a detection algorithm on broadcast data and, as reported by France to an ICAO regional group, found 196 errors above 5 hPa in 747,353 flights at 378 European airports [4, 5]. The full text of that paper could not be obtained for this study.

No published rate was found for the United States, and no study was found that compares business aviation with airlines or one region with another. The United States differs from Europe in two relevant ways. Settings are given in inches of mercury, and the change from the standard setting to the local one is made at 18,000 ft, well before the approach, where in Europe it is made at a few thousand feet [3, 6].

This study uses open data to measure the rate at 119 airports in both regions and to answer two questions. Do business jets land on a wrong setting more often than airline jets at the same airport on the same day? And does the rate differ between Europe and the United States, in particular for landings with the standard setting still selected?

2 Methods

2.1 Data and samples

The adsb.lol archive publishes each day's ADS-B traces under an open licence [7]. Two samples of 30 UTC days in 2026 were used. The first had been drawn at random for an earlier study of approach paths that never used the altimeter-setting field. The second was drawn at random from the 263 eligible days of 2026 not already used; one drawn day (27 March) could not be processed and was replaced by the first reserve day (25 March), as the plan provided. Airports (119, chosen in advance, 76 in the United States and 43 in Europe) and aircraft types (65 business-jet and 58 airline jet types) were fixed lists. Airline-type aircraft were kept only with an airline-format callsign. Airport elevations came from OurAirports [8].

2.2 Setting at landing

A landing is the last airborne point of a track aligned with a runway and followed by ground movement or no later climb. The setting at landing is the most frequent value of the selected barometric pressure setting recorded for the aircraft in the 120 seconds before that point [9]. A landing was excluded if three or more different values were recorded in that window, or if the value was outside 940 to 1060 hPa.

The recorded field mixes two sources. One is the ADS-B broadcast, in steps of 0.8 hPa. The other has 0.1 hPa resolution and is probably a Mode S register decoded by the receiver software. A value that is not on the 0.8 hPa grid can only have come from the second source. Section 3.3 reports results with and without such values.

2.3 Reference

The reference for a landing is the median setting at landing of other aircraft landing at the same airport within 30 minutes either side. It required at least five other aircraft, at least 60% of them within 1.0 hPa of their median. Landings without a reference were excluded.

The reference was checked against the most recent routine or special weather report (METAR) for the airport issued in the 70 minutes before the landing, from the Iowa State University archive [10]. Reports outside 940 to 1060 hPa and isolated spikes were removed

(25 reports). The archive gives the setting in inches of mercury to two decimal places, so European values pass through a rounding of about 0.3 hPa. Palm Beach was missing from the weather data.

2.4 Errors and exclusions

The difference is the setting at landing minus the reference, rounded to 0.1 hPa. Thresholds were 2 hPa (primary), 5 hPa and 10 hPa. Because the broadcast moves in steps of 0.8 hPa, a threshold of 2 hPa against a reference built from other broadcasts is in effect 2.4 hPa. "Standard setting at landing" means a setting within 0.5 hPa of 1013.25 that was at least 5 hPa from the reference, among landings where the reference was at least 5 hPa from standard. Two exclusions were made without regard to class: settings matching station pressure at airports above 300 ft (use of QFE), and all landings by any aircraft that was wrong on at least half of four or more landings.

2.5 Statistical analysis

Business jets were compared with airline jets by the Mantel-Haenszel common odds ratio across strata of airport and UTC date, with a Cochran-Mantel-Haenszel test and a Robins-Breslow-Greenland interval. Because landings by one operator are not independent, intervals were also obtained by resampling whole operators (airline callsign prefix; aircraft without one treated singly) and whole airports. Rates are crude, with intervals from resampling airport-days. Regions were compared by the ratio of crude rates, with a p-value from permuting the region label across airports.

2.6 Pre-registration, deviations and review

The study was pre-registered twice in a public repository (see Materials).

First sample (commit `be41aa0`, 11 November 2026, 13:18 UTC, before any altimeter-setting data from the study days was read). Two hypotheses: H1, that business jets land with a setting 2 hPa or more from the reference more often than airline jets; H2, that landing with the standard setting selected is more frequent in Europe than in the United States. The registered setting at landing was the *last* value in the 120-second window. Review found that measure flawed: on some aircraft, mostly European airliners, the second source shows the standard setting for a few seconds around touchdown after the correct setting had been shown throughout the approach, and a few aircraft broadcast values that jumped between impossible ones. The most-frequent-value measure described above was adopted in response. Appendix A gives the first-sample results exactly as registered.

Second sample (commit `7306d3b`, 11 November 2026, 14:39 UTC, before any of it was downloaded). The corrected measure and four hypotheses were fixed: business jets against airline jets for errors of 2 hPa or more (R1), 5 hPa or more (R2), 2 hPa or more among aircraft with ten or more analysed landings (R3), and the standard setting at landing (R4), with Holm correction. R1 would count as replicated if supported with an odds ratio between 2 and 6. The frozen script was run once after all 30 days were processed.

The confirmatory results in this paper are therefore H1 and H2 on the first sample (Appendix A) and R1 to R4 on the second (Table 3). Everything else is exploratory, including all first-sample results on the corrected measure, the separation of values by source, the operator-level intervals, the cleaning of weather reports and the comparison of regions. Table and text say which sample each figure comes from.

All review of this paper has been by three independent AI reviewers that had not seen the work being produced (see Acknowledgement). "The reviewers" below refers to them.

3 Results

3.1 Samples

The first sample held 809,959 landings and the second 825,926 (Table 1). In the second, no setting was recorded on 24,198 landings; 74 had three or more values in the window, 20 had implausible values, 3 matched QFE, and 53 were by 6 aircraft, all business jets, excluded under the half-wrong rule. With landings lacking a reference also removed, 730,324 were analysed: 667,394 by 17,642 airline aircraft and 62,930 by 11,602 business jets. The first sample gave 713,758 analysed landings (650,245 and 63,513). Far fewer business-jet landings could be analysed. The two samples share most of their aircraft: 99.9% of airline and 96.6% of business-jet landings in the second sample were by aircraft that also appear in the first.

Table 1: Coverage by class and region.

	First sample			Second sample		
Group	Landings	Recorded	Analysed	Landings	Recorded	Analysed
Airline jets, Europe	275,237	99.6%	93.6%	282,316	99.6%	93.9%
Airline jets, United States	439,679	97.2%	89.3%	450,250	97.2%	89.3%
Business jets, Europe	17,416	97.9%	74.9%	16,821	97.8%	75.1%
Business jets, United States	77,627	86.9%	65.0%	76,539	86.9%	65.7%

3.2 How often the setting was wrong

In the second sample the setting differed from the reference by 2 hPa or more on 2.04 per 1,000 landings (95% interval 1.85 to 2.25), by 5 hPa or more on 0.19 (0.16 to 0.23), and by 10 hPa or more on 0.060 (0.043 to 0.079). The first sample gave 1.98, 0.20 and 0.060. Table 2 and Figure 1 give the rates by class and region. Nine in ten of the discrepancies of 2 hPa or more were smaller than 5 hPa, most of them 2.4 or 3.2 hPa.

Table 2: Landings per 1,000 with the setting wrong by at least the amount shown. Crude rates, first sample / second sample. "Grid only" rows keep only values on the 0.8 hPa broadcast grid (Section 3.3); US rows are unchanged by that restriction.

	2 hPa	5 hPa	10 hPa
Airline jets, Europe	0.26 / 0.31	0.10 / 0.11	0.023 / 0.041
grid only	0.19 / 0.21	0.070 / 0.045	0.012 / 0.015
Airline jets, United States	2.05 / 2.03	0.14 / 0.10	0.038 / 0.030
Business jets, Europe	1.69 / 1.98	1.30 / 1.19	0.54 / 0.47
grid only	1.08 / 1.51	0.69 / 0.72	0.46 / 0.40
Business jets, United States	10.3 / 11.3	0.87 / 1.07	0.30 / 0.30
All landings	1.98 / 2.04	0.20 / 0.19	0.060 / 0.060
grid only	1.94 / 2.00	0.18 / 0.16	0.055 / 0.049

3.3 Values from the second source

Values off the 0.8 hPa grid were 562 of 277,801 analysed European landings in the second sample (0.2%) and 8 in the United States, but they were 23 of the 44 European errors of

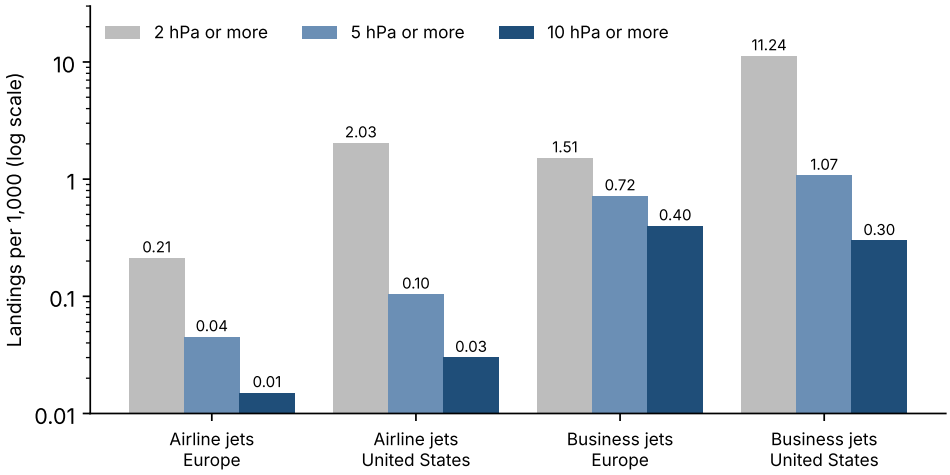


Figure 1: Landings per 1,000 with the setting wrong by at least the amount shown. Second sample, values on the broadcast grid, crude rates, logarithmic scale.

5 hPa or more (17 of 44 in the first sample). The reviewers traced them to a small number of older aircraft, among them A300-600 freighters and 737 Classics, that show a value at or near 1013 on most or all of their landings and made too few landings to trip the half-wrong rule. These are far more likely to be a property of the avionics or the decoding than crews landing on the standard setting. European rates are therefore given both ways in Table 2, and the grid-only figures are the better estimate. The grid test is approximate, since one 0.1 hPa value in eight falls on the grid by chance.

The European rates on grid values (0.2 to 0.3 per 1,000 at 2 hPa and 0.03 at 10 hPa) are well below the two earlier European studies: about 1.3 and 0.23 in the French study [3], and 0.26 at 5 hPa reported for EUROCONTROL [5] against 0.08 to 0.10 here. The French study used the radar downlink, which may be the same second source, but this was not tested.

3.4 Business jets against airline jets

Within the same airport and date, business jets had about four times the odds of airline jets of landing with a setting 2 hPa or more from the reference, and about fourteen times the odds for 5 hPa or more (Table 3). All four pre-registered hypotheses on the second sample were supported after Holm correction (each $p < .001$), and R1 met its replication criterion. The first-sample hypothesis H1, on the registered measure, was also supported (odds ratio 3.42; Appendix A).

Table 3: Business jets against airline jets within airport and date. Mantel-Haenszel odds ratios; business-jet and airline events in informative strata in square brackets. The first interval is the Robins-Breslow-Greenland one; the second, for the second sample, comes from resampling whole operators and is the one to rely on. Second-sample tests were pre-registered; first-sample figures are exploratory.

Outcome	First sample	Second sample
R1: wrong by 2 hPa or more	3.5 (3.0–4.0) [342, 856]	4.0 (3.5–4.7; 2.7–5.4) [377, 877]
R2: wrong by 5 hPa or more	8.6 (5.2–14.2) [49, 79]	13.9 (8.9–21.8; 7.9–23.7) [64, 63]
R3: as R1, aircraft with 10+ landings	2.6 (2.1–3.1) [130, 797]	2.6 (2.1–3.2; 1.7–3.6) [135, 830]
R4: 1013 at landing, 5 hPa+ away	32 (13–77) [27, 19]	35 (16–78; 12–114) [34, 18]

Crude rates. Over all airports the rates of a discrepancy of 2 hPa or more were 9.4 per 1,000 for business jets and 1.35 for airline jets in the second sample. The crude ratio is larger than the odds ratio because business jets land more at airports where rates are higher for every class.

Robustness. In the second sample the R1 odds ratio was 3.9 in the United States and 7.4 (4.3 to 13.0) in Europe, 3.8 with the hour added to the strata, 4.8 when the reference needed ten or more aircraft with over 95% agreement, and 4.0 on grid values only (R2 14.4, R3 2.6). Resampling whole airports gave 3.1 to 5.4. In the first sample it held under every sensitivity analysis run (3.1 to 5.1), and the reviewers confirmed it with conditional and fixed-effects logistic models, with the reference built from airline aircraft only (3.7), and with their own independent code on the second sample (4.0).

By size of error. In the second sample the odds ratio was 3.4 (2.9 to 4.0) for discrepancies of 2 to 5 hPa and 20 (9 to 44) for 10 hPa or more, on 19 and 21 events; in the first, 3.1 and 16.8 (6.9 to 41.2). The larger ratios are imprecise. Five of the 21 business-jet errors of 10 hPa or more in the second sample, and five of 22 in the first, were non-standard values 30 hPa or more from the reference, against none and one for airline jets; the reviewers inspected the traces and judged four in the second sample to be corrupt broadcasts. Excluding all such values, R2 is 13.0 (8.2 to 20.5) and the 10 hPa odds ratio 15.8 (7.0 to 35.9).

By how often the aircraft was seen. The gap narrows among aircraft that landed often in the sample (Table 4). Among aircraft with twenty or more landings the odds ratio was 2.2 (1.6 to 3.0) in the second sample and 2.0 in the first. Of business-jet errors of 10 hPa or more, 15 of 21 in the second sample and 19 of 22 in the first were by aircraft with fewer than ten landings.

Table 4: Discrepancies of 2 hPa or more per 1,000 landings, by the number of landings the aircraft made in the sample (first sample / second sample).

Landings by the aircraft	1	2–3	4–9	10–19	20 or more
Business jets	13.5 / 19.9	10.8 / 14.4	10.4 / 9.1	7.2 / 8.0	5.2 / 6.4
Airline jets	16.4 / 11.7	3.0 / 6.0	1.5 / 1.3	1.5 / 1.4	1.3 / 1.3

By operator, type and airport (first sample). Two fractional-ownership operators flew 26% of analysed business-jet landings. Among the 60 airline callsign groups with at least 1,500 landings, rates ran from 0 to 14 per 1,000 with a median of 0.5; several large carriers had none. Regional jets were highest among airline types (2.6 to 5.1 per 1,000). Business-jet types with at least 1,000 landings ran from 3.1 to 16.2. The spread between operators is as wide as the gap between the classes. Rates for both classes rose with the share of business jets at the airport, where the reference is also noisier. The reviewers found that business jets with an error in the first sample had a higher rate in the second (14.3 against 8.5 per 1,000).

3.5 Standard setting at landing

About half of business-jet errors of 5 hPa or more had 1013 selected (35 of 69 in the second sample, 32 of 61 in the first). Table 5 gives the counts. Within airport and date, business jets had far higher odds than airline jets (R4: 35, operator-level interval 12 to 114). Sixteen of the 18 European airline events in the second sample were second-source values; on grid values only the odds ratio is 74 (21 to 257), on 28 business-jet and 7 airline events.

The reviewers examined the traces of all 35 business-jet events in the second sample. On 28 the value was 1013 throughout the observed part of the approach, which was the last four

Table 5: Landings with 1013 selected and the reference at least 5 hPa away: events / eligible landings. The last column keeps only values on the broadcast grid, second sample.

Group	First sample	Second sample	Second, grid only
Airline jets, Europe	15 / 158,123	18 / 172,829	2 / 172,519
Airline jets, United States	6 / 218,843	5 / 216,957	5 / 216,956
Business jets, Europe	14 / 8,040	10 / 7,563	4 / 7,517
Business jets, United States	18 / 27,437	25 / 27,873	25 / 27,871

to ten minutes; five showed a change to 1013 during it. All were one-off for the aircraft, and they occurred across 22 aircraft types, old and recent. They are not stuck transponders and not the touchdown artefact, but the traces cannot show whether the crew's own altimeter was also at 1013.

The first-sample hypothesis H2, that the standard setting at landing is more frequent in Europe, was not supported as registered (rate ratio 1.97, $p = .072$), and the apparent European excess was the touchdown artefact: on the corrected measure the ratio was 0.95 ($p = .85$).

3.6 The United States against Europe

Discrepancies of 2 to 5 hPa were more than ten times as frequent in the United States as in Europe (Table 2). For errors of 5 hPa or more, on all values the United States rate was 1.3 times the European one in the second sample (0.21 against 0.16 per 1,000; $p = .22$). On grid values only it was 2.8 times (0.21 against 0.076; resampling airports, 1.7 to 5.3; $p < .001$), and 2.2 times in the first sample ($p = .002$). The comparison is exploratory and is not adjusted for the mix of traffic.

3.7 The reference against weather reports

The consensus reference was within 1.0 hPa of the latest weather report for 97.5% of 724,711 landings in the second sample (97.2% in Europe, 97.6% in the United States) and within 1.7 hPa for 99.9%; the first sample gave 97.4% and 99.9%. Agreement was lower at airports where business jets were over 60% of traffic (94.3%).

With the weather report as the reference in place of the consensus, which brings in 17,000 to 18,000 business-jet landings per sample that had too little other traffic for a consensus, the odds ratio in the second sample was 4.0 (3.4 to 4.7) at the matched threshold of 2.4 hPa and 13.1 (8.9 to 19.3) at 5 hPa; in the first, 3.5 and 11.5. The two references give the same answer. The frozen script, which did not remove corrupt reports and used a nominal 2.0 hPa threshold that admits two-step differences, gave 2.8 and 9.4. Measured against weather reports, landings without a consensus reference had higher discrepancy rates in both classes (business jets 9.6 against 7.4 per 1,000, airline jets 2.4 against 0.9), so the rates in Table 2 are probably too low.

3.8 Other observations

- **Small discrepancies in the United States.** In the first sample, 74% for airline jets and 86% for business jets had been held unchanged for five minutes or more. Airline ones bunched in the first ten minutes of each hour (177, against 101 to 127 in the other ten-minute periods), which fits the hourly weather update; business-jet ones were flat

(74 to 87). In the second sample only 24% of airline and 11% of business-jet small discrepancies matched a setting published for the destination in the previous two hours.

- **Direction (first sample).** Errors of 2 hPa or more were evenly split. Of errors of 5 hPa or more, 50 were set too high, which puts the aircraft lower than indicated, and 94 too low.
- **Repeat errors (first sample).** Errors were rarely repeated by the same aircraft. The reviewers found that nearly all business-jet errors came from aircraft that broadcast different values on other flights and held one steady value through the approach.
- **Paris.** Le Bourget had too few analysable landings (873 in the first sample) to test the French threefold difference.

4 Discussion

What is established. Compared at the same airport on the same day, the odds that a business jet lands with a broadcast altimeter setting 2 hPa or more from that of other aircraft are about three and a half to four times those for an airline jet. The result was confirmed in a second pre-registered sample of new days, is the same when the airport weather report is the reference, and holds with allowance for landings by the same operator (2.7 to 5.4). It is not produced by a few faulty transponders.

Most of it is small. Nine in ten of the discrepancies are between 2 and 5 hPa, mostly 2.4 or 3.2 hPa, equivalent to about 65 to 90 ft. What they are is not established. The bunching of airline ones after the hourly update points to settings not yet renewed, but only a minority in either class matched a recent setting for the destination. A setting taken from another station is one possible explanation and was not tested; a setting from a station within 100 NM is legitimate en route. These are best described as discrepancies from the local setting, not errors in the sense of the Paris incident.

The large errors are few. Errors of 5 hPa or more occurred on about 1 business-jet landing per 1,000 against about 0.1 for airline jets, in both samples. The ratio is large and was confirmed, and it is the safety-relevant finding. In each sample it rests on 60 to 70 business-jet events, about half of them with 1013 selected. At 10 hPa most came from aircraft seen fewer than ten times, and about a quarter look like corrupt values.

The broadcast may not be the altimeter the crew flew on. The standards define the field as the setting selected by the crew but not which side it comes from [11]. One reviewer, asked to take an avionics perspective, noted that on some light jets flown by one pilot the transponder may take its setting from the unused side, and that such aircraft often lack a warning when the two sides disagree. The data cannot exclude that reading, and the spread of 1013 events across 22 types gives it no clear support either. On this reading some business-jet "standard at landing" events are an unset second altimeter and not a crew flying on the wrong setting. That would still mean no cross-check between the two sides was made.

The United States against Europe. Once the second-source values are set aside, errors of 5 hPa or more were two to three times as frequent in the United States as in Europe, in both samples. This does not support treating the problem as a European one that the 18,000 ft transition altitude prevents. The comparison is exploratory and crude, and the European rates here are well below both earlier studies, which may have included the same second-source values.

Consequences. A 10 hPa error is about 280 ft. On an approach with ground-based vertical guidance the glidepath is unaffected, but the decision altitude, the altitude check on the glidepath, step-down and missed-approach altitudes are all displaced. On an approach using barometric vertical guidance the whole path is displaced [1]. Radio altimeters, terrain warning using satellite height and controller warnings remain as defences. The study does not know which type of approach was flown.

5 Limitations

- **Review.** The paper has been reviewed only by AI reviewers. Some figures attributed to the reviewers were not re-derived by the author's analysis; `AUDIT.md` lists which. The pre-registrations, analysis and review all took place within two days, and the commit times are not independently certified.
- **The measure is not direct.** The recorded value combines two sources that can disagree, and which altimeter either reflects is unknown. Values from the second source could be separated only approximately. The measure was chosen after seeing a problem in the first sample, then fixed in advance for the second.
- **Inferred reference.** The reference is the consensus of other aircraft. A wrong value given to every aircraft would not be detected. It agreed with weather reports within 1 hPa for 97% of landings.
- **Coverage and selection.** Only 65% of US and 75% of European business-jet landings could be analysed, against about 90% for airline jets. The missing landings are aircraft that do not broadcast a setting, which is optional in the United States, and landings with too little other traffic. The reviewers found that business-jet types with lower broadcast coverage had higher error rates among the aircraft that did broadcast, so the direction of any bias from non-broadcasting aircraft is unknown.
- **Exposure.** Airline aircraft averaged about 37 landings per sample and business jets about 5.5. The per-landing comparison is dominated by rarely seen business jets, and the gap is smaller among frequently seen aircraft.
- **Lag counted as discrepancy.** A crew holding a correct but older value is counted at the 2 hPa threshold.
- **Classes.** The airline class includes cargo and some military and government flights with airline-format callsigns. Operator type is inferred from aircraft type and callsign. Test and training flights are included.
- **No operational context.** Approach type, weather, minima, crew composition and what ATC said are unknown.
- **Sample.** Sixty days in 2026 at 119 airports chosen by judgement. The second sample has new days but the same airports and mostly the same aircraft. It confirms that the result is stable over time and was not produced by choices made after seeing the data. It does not show that the result would hold for other airports or fleets.
- **Prior work not fully read.** The EUROCONTROL paper was seen only as an abstract and through secondary reports, and its figures quoted here come from a secondary source. It may contain breakdowns that this paper describes as not found. No systematic search was made for unpublished monitoring by air navigation service providers.

6 Conclusion

In open ADS-B data from 119 airports, a broadcast altimeter setting at least 2 hPa from the local one was present on about 2 landings per 1,000, and one at least 5 hPa away on about 2 per 10,000. Business jets had about four times the odds of airline jets of the first and about fourteen times the odds of the second, at the same airport on the same day, a result that was predicted in advance and confirmed on a second sample. Large errors were not confined to Europe. Whether the business-jet excess reflects the altimeter the crew flew on, a second altimeter left unset, or differences in how settings are obtained cannot be settled from broadcast data, and needs avionics evidence for the main types and operator flight data.

A First-sample results as pre-registered

These results use the registered measure, the last value in the 120-second window, and the frozen code, whose hashes match the pre-registration. They are superseded by the main text for the reasons given in Section 2.6.

Sample. A setting was recorded in the last two minutes on 97.0% of the 809,959 landings. After exclusions (27 implausible values, 3 landings on QFE, and 89 landings by 9 aircraft, 8 of them business jets, under the half-wrong rule), 713,835 landings were analysed: 650,307 by 17,617 airline aircraft and 63,528 by 11,618 business jets. Stratum dates span 31 UTC dates because three landings fell just after midnight.

H1. Within the same airport and date, the odds ratio of an error of 2 hPa or more for business jets against airline jets was 3.42 (95% interval 2.94 to 3.98; resampling airports, 2.77 to 4.27; Holm-adjusted $p < .001$; 684 informative strata). The crude rates were 8.53 per 1,000 (542 of 63,528) and 1.43 per 1,000 (930 of 650,307). H1 is supported.

H2. "Standard left set" was registered as a setting within 0.5 hPa of 1013.25 that differed from the reference by 2 hPa or more, among landings where the reference was at least 3 hPa from standard. It was recorded on 71 of 188,811 eligible landings in Europe (0.38 per 1,000) and 58 of 303,311 in the United States (0.19 per 1,000): rate ratio 1.97 (0.99 to 3.49; permutation $p = .072$). H2 is not supported.

Rates. The registered measure gave 2.06 errors of 2 hPa or more per 1,000 landings, 0.26 of 5 hPa or more and 0.083 of 10 hPa or more.

Effect of the correction. The corrected setting differed from the registered one on 1,784 first-sample landings, and 80 landings were excluded as unstable. On the corrected measure the H1 odds ratio was 3.46 (2.97 to 4.02).

Other deviations. The rate tables by aircraft type use lower cut-offs than the frozen script. In the frozen replication script the two rate labels in the United States against Europe output are transposed; the ratio and p-value are unaffected. One trace file on the reserve day of the second sample could not be read.

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findings changed the paper and are recorded in `AUDIT.md`. The author is responsible for the final text.

Competing interests. The author is the founder of CitadelAero, which builds AI tools for aviation, and is employed as a business-jet pilot.

Disclaimer. The views expressed are the author's own and do not represent those of any employer or regulator. Nothing here is operational, legal or regulatory advice.

Materials. The pre-registration, code, results and audit record for this paper are at `https://github.com/JamesLeCuirot/Research/tree/prereg/08-wrong-altimeter-setting/papers/08-wrong-altimeter-setting`.

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